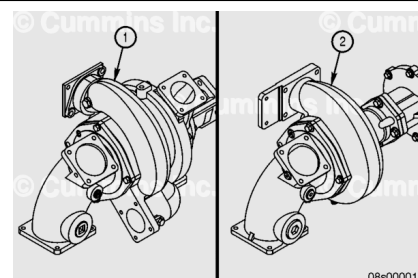


Trainings ▾

General Information

This procedure is used for engines equipped with the dual impeller water pump (1).

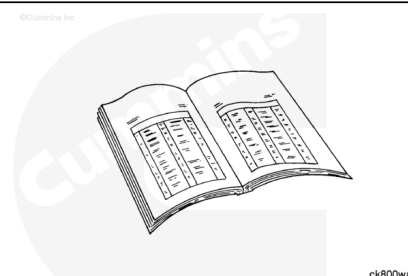
For engines equipped with the single impeller, jacket water pump (2), use the following procedure. Refer to Procedure 008-005 (/qs3/pubsys2/xml/en/procedures/28/28-008-005-tr.html) in Section 8.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

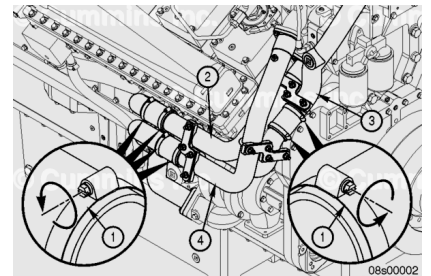
- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13.

(/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)

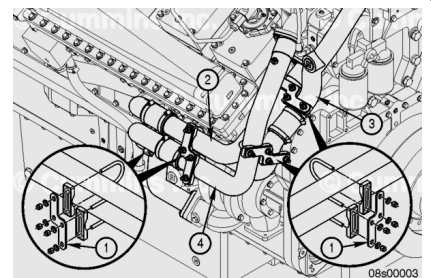
Remove

Loosen the six hose clamps (1) from the water bypass (2), LTA bypass (3), and the LTA supply (4) pipe connection hoses.



Remove the four sets of brackets, U-bolt clamps, and wear pads (1) from the water bypass (2), LTA bypass (3), and the LTA supply (4) tubes.

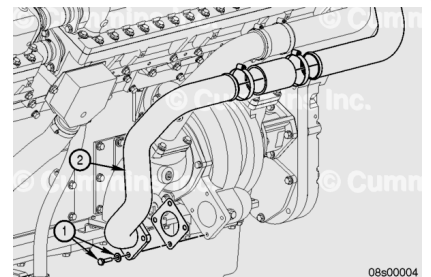
Discard the nylock nuts.



Remove the four capscrews and lock washers (1) that secure the lower LTA supply tube section (2) to the water pump.

Remove the lower LTA pipe section, connection hose, clamps and gasket.

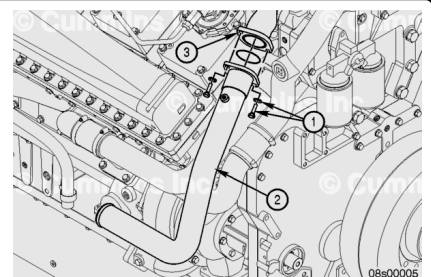
Discard the gasket.



Remove the two capscrews and lock washers (1) that secure the upper LTA supply tube section (2) to the connection block (3).

Remove the upper LTA supply tube section (2) and o-ring from the connection block.

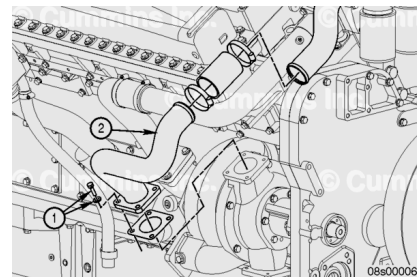
Discard the o-ring.



Remove the four capscrews and lock washers (1) that secure the lower LTA bypass tube section (2) to the water pump.

Remove the lower LTA bypass tube section (2), connection hose, clamps, and gasket.

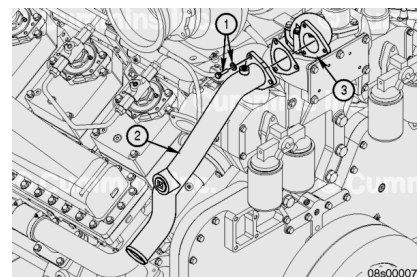
Discard the gasket.



Remove the three capscrews and lock washers (1) that secure the upper LTA bypass tube section (2) to the thermostat connection (3).

Remove the upper LTA bypass tube section (2) and gasket from the thermostat connection.

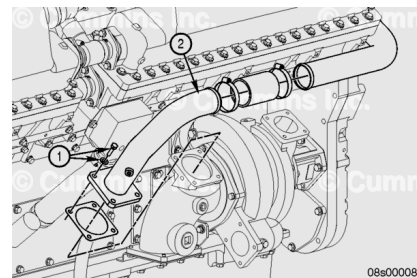
Discard the gasket.



Remove the four capscrews and lock washers (1) that secure the lower jacket water bypass tube section (2) to the water pump connection.

Remove the lower water bypass tube section (2), connection hose, clamps, and gasket.

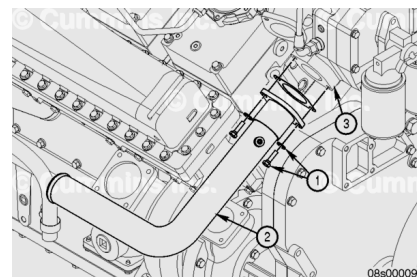
Discard the gasket.



Remove the two capscrews and lock washers (1) that secure the upper jacket water bypass tube section (2) to the thermostat housing (3).

Remove the upper jacket water bypass tube section (2) and gasket from the thermostat housing.

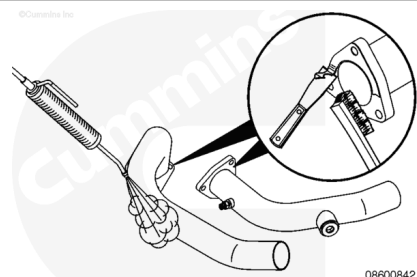
Discard the gasket.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing, to prevent personal injury when using a steam cleaner or high-pressure water.



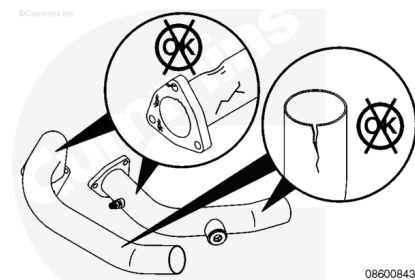
Use a gasket scraper and wire brush to clean the gasket sealing surfaces.

Use steam to clean the tubes.

Check the tube gasket sealing face and tube for damage.

Replace the tube if it is cracked or otherwise damaged.

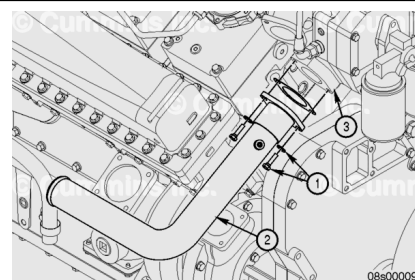
Replace the connection hose and hose clamps.



Install

Install the gasket and upper jacket water bypass tube section (2) to the thermostat housing (3) using the two capscrews and lock washers (1).

Hand tighten the capscrews.



Install the connection hose and 2 clamps onto the upper jacket water bypass tube (2), leaving the clamps loose.

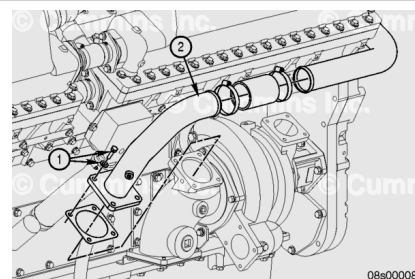
Install the end of the lower jacket water bypass tube (2) into the connection hose.

Install the gasket and lower jacket water bypass tube (2) onto the water pump inlet connection using the 4 capscrews and lock washers (1).

Hand tighten the capscrews.

Align the upper and lower tube sections in the connection hose.

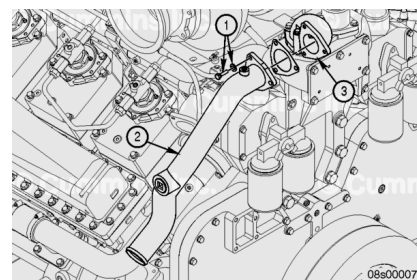
Tighten the four capscrews on the water pump and two capscrews on the thermostat housing.



Torque Value: 45 n•m [33 ft-lb]

Install the gasket and upper LTA bypass tube section (2) to the thermostat connection (3) using the three capscrews and washers (1).

Hand tighten the capscrews.



Install the connection hose and two clamps onto the upper LTA bypass tube, leaving the clamps loose.

Install the end of the lower LTA water bypass tube (2) in the connection hose.

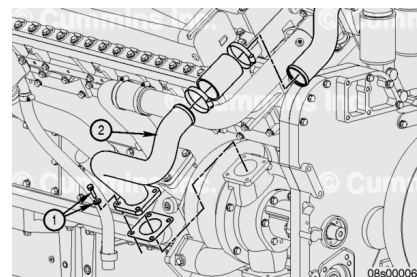
Install the gasket and lower LTA water bypass tube (2) to the water pump using the four capscrews and lock washers (1).

Hand tighten the capscrews.

Align the upper and lower tube sections in the connection hose.

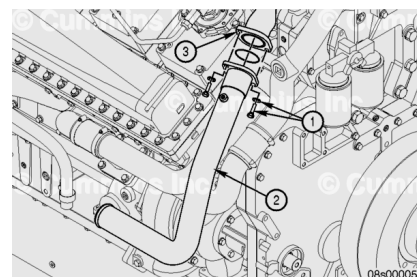
Tighten the four capscrews on the water pump and three capscrews on the thermostat connection.

Torque Value: 45 n•m [33 ft-lb]



Install the o-ring and upper LTA supply tube section (2) to the connection block (3) using the two capscrews and washers (1).

Hand tighten the capscrews.



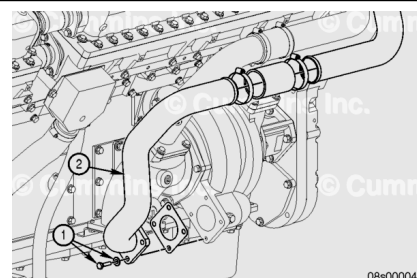
Install the connection hose and two clamps onto the upper LTA supply tube, leaving the clamps loose.

Install the end of the lower LTA water bypass tube (2) in the connection hose.

Install the gasket and lower LTA water bypass tube (2) to the water pump using the four capscrews and lock washers (1).

Hand tighten the capscrews.

Align the upper and lower tube sections in the connection hose.



Tighten the two capscrews on the upper LTA supply tube.
Tighten the four capscrews on the lower LTA water bypass tube (2).

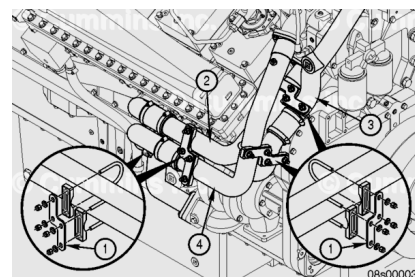
Torque Value: 45 n•m [33 ft-lb]

Install the four sets of brackets, U-bolt clamps, and wear pads (1) to the water bypass (2) , LTA bypass (3) and LTA supply (4) pipes.

Note : Be sure the lower LTA supply to LTA bypass tube bracket is installed below the welds on the tubes.

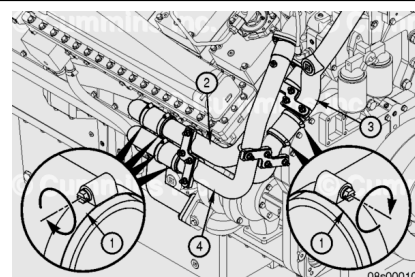
Alternately tighten the clamp nuts and be sure the bracket pulls down evenly.

Torque Value: 6 n•m [53 in-lb]



Tighten the six connection hose clamps (1) on the jacket water bypass (2), LTA bypass (3), and LTA supply (4) tube connection hoses.

Torque Value: 6 n•m [53 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the battery cables. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine to normal operating temperature and pressure and check for leaks.

Last Modified: 24-Sep-2015
